

DRY-RUN GUIDE | avg_matrix.py

Source: Python\3rd_Belt\Matrix\avg_matrix.py

CORE BEHAVIOR

The helper sums the cells and divides by rows*cols, but those names are local to main and are not visible in the helper. A non-empty input reaches `NameError`.

REPRESENTATIVE INPUT

2x2 rows '1 2' and '3 4'.

STEP TRACE

total_sum becomes 10; evaluating rows*cols in average_of_matrix fails because rows is undefined in that scope.

EXPECTED OUTPUT / RETURN

No average is printed; `NameError ('rows' is not defined)`.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt\Matrix\avg_matrix.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Compute the average value of all elements in a matrix.
002 | # Add your solution here.
003 |
004 |
005 | def average_of_matrix(matrix):
006 |     if not matrix or not matrix[0]:
007 |         return 0.0
008 |
009 |     total_sum = sum(sum(row) for row in matrix)
010 |     total_elements = rows*cols
011 |     return total_sum / total_elements
012 |
013 |
014 | if __name__ == "__main__":
015 |     rows, cols = map(int, input().split())
016 |     matrix = [list(map(int, input().split())) for _ in range(rows)]
017 |     print(f"{average_of_matrix(matrix):.1f}")
018 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.